

Polarity-Routed Cascade Diffusion for Crop Waste Prediction in U.S. Agricultural Economic Networks

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Abstract

Crop waste costs U.S. agriculture roughly \$19 billion annually in insurance indemnities alone. Existing prediction models treat counties as independent units, ignoring the economic contagion that propagates through commodity supply chains. We propose polarity-routed cascade diffusion, a feature engineering framework where negative price shocks propagate through commodity-specific supply chain graphs and positive signals follow geographic proximity networks. The core insight: oversupply cascades spread along commodity channels (a corn glut in Iowa depresses corn prices in Ohio), while demand-driven gains diffuse through regional geography. We validate on 638,622 USDA records spanning 2,130 U.S. counties and three major commodities (corn, soybeans, wheat) from 2015 to 2025. On a random split, the polarity-routed model achieves 0.935 AUC-ROC—a significant improvement over the local-only baseline at 0.917 ($p < 0.001$) and over symmetric diffusion at 0.934 ($p = 0.0002$). The entire framework uses a 13,703-parameter multi-layer perceptron; novelty lies in the features, not the classifier. A temporal split (train 2015–2019, test 2022–2024) reveals that the local-only model (0.910 AUC-ROC) dominates all graph-augmented variants, exposing the non-stationarity of economic contagion patterns across time. We ground these empirical findings in a formal economic model showing that waste contagion arises from market connectivity and that positive and negative supply shocks have provably asymmetric effects on neighboring regions. These results carry practical implications for crop insurance pricing and county-level early warning systems.

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1. Introduction

In 2023, the USDA Risk Management Agency disbursed over \$19 billion in crop insurance indemnities (USDA Risk Management Agency, 2024). Much of that money covered crops that grew just fine—then went to waste. Fields were abandoned because market prices fell below harvesting costs. Standing corn was plowed under because a bumper harvest three states away flooded the commodity market. Globally, food loss and waste constitute a trillion-dollar drag on agricultural economies (Food and Agriculture Organization of the United Nations, 2019).

These waste events are not independent. They cascade.

When Iowa produces record corn yields, the resulting supply glut depresses prices across the Midwest. Farmers in Illinois, Indiana, and Ohio—whose crops may be perfectly healthy—face the cold arithmetic of harvesting at a loss. Some choose to abandon. This mechanism mirrors financial contagion: a shock at one node propagates through network connections to cause failures at distant nodes (Acemoglu et al., 2015; Elliott et al., 2014). The analogy runs deep. Agricultural markets share the structural properties that produce cascading failures in financial networks: high interconnectedness through commodity exchanges, threshold effects where small price changes tip harvesting decisions, and asymmetric responses to positive versus negative shocks (Battiston et al., 2012).

Yet every crop prediction model treats counties as islands. Hundreds of papers use satellite imagery, weather data, and soil characteristics to forecast yields for individual regions (van Klompenburg et al., 2020; Kang et al., 2020; Basso and Liu, 2019; Lobell et al., 2011). These models predict the wrong thing. They predict how much a crop *will produce*, not whether it *will be wasted*. High-yield crops go entirely to waste when prices crash. Low-yield crops get fully utilized when prices surge. Waste depends on biology *and* economics—and the economics propagate across regions.

Graph neural networks offer a natural framework for modeling inter-regional dependencies (Kipf and Welling, 2017; Veličković et al., 2018). However, they bring significant complexity: end-to-end training over graph structures, poor scalability to large networks, and reduced interpretability. Recent

work on simplified graph convolutions (Wu et al., 2019; Rossi et al., 2020) showed that pre-computing graph features—propagating node signals through the adjacency matrix before training—can match GNN performance at a fraction of the cost.

We build on this insight with a critical domain-specific innovation: **polarity-routed diffusion**. Standard graph diffusion propagates all signals through a single topology. We argue—and demonstrate—that economic shocks in agriculture are *polarity-dependent*. Negative price shocks (from oversupply) propagate through commodity-specific networks: a corn glut in Iowa affects other corn-producing counties. Positive price signals (from undersupply or demand spikes) propagate through geographic proximity: when prices rise regionally, nearby counties benefit regardless of commodity mix. Routing signals through different graph topologies based on their polarity captures this fundamental asymmetry.

We formalize this intuition with a simple economic model (Section 2) that yields three testable predictions. Supply shocks in one county depress prices—and increase waste risk—in market-connected counties. This contagion is provably asymmetric: positive and negative shocks follow different propagation channels. These theoretical results map directly onto our feature engineering choices.

Our contributions:

1. **Polarity-routed cascade diffusion.** We introduce a feature engineering framework where negative and positive economic signals propagate through different graph topologies (commodity-specific vs. geographic) at multiple hops. To our knowledge, this is the first work to route graph signals through different topologies based on shock polarity.
2. **Formal economic model of waste contagion.** We derive propositions showing that waste contagion arises from market connectivity and that positive and negative supply shocks have provably asymmetric effects. These results justify the polarity routing mechanism from first principles.
3. **Cascade decay signatures and cross-commodity interference.** We introduce hop-by-hop attenuation ratios that capture shock persistence and cross-commodity features that measure interference from other commodities grown in the same county.
4. **Large-scale empirical validation.** We evaluate on 638,622 samples across 2,130 U.S. counties and three commodities (2015–2025), with both

random and temporal splits. The temporal split reveals important non-stationarity in spatial economic patterns.

5. Computational efficiency by design. The entire framework—pre-computed features fed into a 13,703-parameter MLP—trains in under a minute. The novelty is in the features, not the classifier.

2. Economic Model of Crop Waste Contagion

Why does a bumper harvest in one county cause crop waste hundreds of miles away? Standard agronomic models treat waste as a local phenomenon—drought, pest damage, frost—and ignore the market channel entirely. We formalize the economic mechanism that propagates waste across regions, derive testable predictions, and show how these predictions map to specific feature engineering choices in our cascade diffusion framework.

2.1. Model Setup

Consider N agricultural regions (counties) indexed by $i = 1, \dots, N$. Each region i grows a commodity with production quantity q_i , determined by biophysical conditions (weather, soil, planted acreage), and faces a harvesting cost $c_i > 0$ encompassing labor, fuel, and equipment. The farmer in region i observes a local market price p_i and harvests if and only if revenue exceeds cost:

$$p_i \cdot q_i > c_i. \quad (1)$$

When this condition fails, the standing crop is abandoned—left unharvested and wasted.

The local price depends on aggregate regional supply through a linear inverse demand system:

$$p_i = P^{\text{global}} - \sum_{j=1}^N \beta_{ij} q_j, \quad (2)$$

where $P^{\text{global}} > 0$ is the national or world reference price and $\beta_{ij} \geq 0$ captures *market connectivity* between regions i and j .

95 *Interpretation of β_{ij} .* The coefficient β_{ij} encodes how strongly production in
 96 region j depresses the price received by farmers in region i . Four properties
 97 govern its structure:

- 98 (i) $\beta_{ij} > 0$ if and only if regions i and j compete in the same downstream
 99 market (shared processors, packers, or distribution channels).
- 100 (ii) β_{ij} increases with geographic proximity, since transportation costs de-
 101 termine which buyers are accessible.
- 102 (iii) β_{ij} increases with commodity similarity—two adjacent corn-producing
 103 counties interact more strongly than a corn county and a soybean county.
- 104 (iv) $\beta_{ij} \approx 0$ for distant or market-disconnected regions.

105 These properties make the matrix $\mathbf{B} = [\beta_{ij}]$ sparse, spatially structured, and
 106 commodity-dependent—precisely the adjacency structure our graph model
 107 learns from data.

108 *Waste probability.* Define the *profit margin* for region i as

$$\pi_i = p_i \cdot q_i - c_i = \left(P^{\text{global}} - \sum_j \beta_{ij} q_j \right) q_i - c_i. \quad (3)$$

109 Region i wastes its crop whenever $\pi_i \leq 0$. In a stochastic setting where q_j
 110 is drawn from a distribution F_j (reflecting weather uncertainty), the waste
 111 probability is

$$\Pr(\text{waste}_i) = \Pr(\pi_i \leq 0) = \Pr\left(\sum_j \beta_{ij} q_j \geq P^{\text{global}} - \frac{c_i}{q_i}\right). \quad (4)$$

112 2.2. Waste Contagion

113 **Proposition 1** (Waste Contagion). *Let region j experience a positive supply*
 114 *shock: q_j increases to $q_j + \Delta q_j$ with $\Delta q_j > 0$, while all other quantities $\{q_k\}_{k \neq j}$*
 115 *remain unchanged. If $\beta_{ij} > 0$, then the waste probability in region i strictly*
 116 *increases, even when region i 's own biophysical conditions are unchanged.*

117 *Proof.* From Equation (2), the price in region i after the shock is

$$p'_i = P^{\text{global}} - \sum_{k \neq j} \beta_{ik} q_k - \beta_{ij}(q_j + \Delta q_j) = p_i - \beta_{ij} \Delta q_j.$$

118 Since $\beta_{ij} > 0$ and $\Delta q_j > 0$, we have $p'_i < p_i$. The post-shock profit margin
 119 becomes

$$\pi'_i = p'_i \cdot q_i - c_i = \pi_i - \beta_{ij} \Delta q_j \cdot q_i < \pi_i.$$

120 Because $q_i > 0$ (region i has a standing crop), the margin drops by the strictly
 121 positive quantity $\beta_{ij} \Delta q_j q_i$. Hence $\Pr(\pi'_i \leq 0) > \Pr(\pi_i \leq 0)$, and the waste
 122 probability in region i strictly increases. $\square$

123 Proposition 1 captures the central paradox of crop waste: good weather in
 124 one county can destroy value in another. The mechanism is purely economic—
 125 no pathogen spreads, no pest migrates. A bumper harvest floods shared
 126 markets, crashes the local price, and pushes marginal farmers below their
 127 break-even threshold. The strength of this contagion depends entirely on β_{ij} ,
 128 the market connectivity (Acemoglu et al., 2015).

129 **Corollary 1** (Asymmetric Propagation). *Positive and negative supply shocks*
 130 *in region j have asymmetric effects on waste in region i :*

131 (a) **Bumper harvest** ($\Delta q_j > 0$): p_i falls by $\beta_{ij} \Delta q_j$, increasing waste
 132 probability in i .

133 (b) **Crop failure** ($\Delta q_j < 0$): p_i rises by $\beta_{ij} |\Delta q_j|$, decreasing waste proba-
 134 bility in i .

135 *Consequently, the effect of neighbor shocks on waste risk is directionally*
 136 *asymmetric: positive supply shocks are harmful and negative supply shocks*
 137 *are beneficial to region i 's harvest viability.*

138 The proof follows immediately from Proposition 1 by substituting $\Delta q_j < 0$.
 139 This asymmetry has a direct implication for feature engineering: a predictive
 140 model must condition on the *sign* of neighbor shocks, not merely their
 141 magnitude. Symmetric diffusion—treating $+\Delta q_j$ and $-\Delta q_j$ identically—
 142 discards the directional information that distinguishes waste-inducing from
 143 waste-relieving shocks. This motivates the polarity routing in our cascade
 144 diffusion framework.

145 2.3. Testable Hypotheses

146 The model generates three predictions that map directly to our ablation
 147 experiments.

148 **Hypothesis 1** (Graph Structure Improves Prediction). *Because waste con-*
149 *tagion propagates through the connectivity matrix $\mathbf{B}$, models that capture*
150 *inter-regional dependencies will outperform models that treat each region*
151 *independently.*

152 *Test.* Compare polarity-routed (which includes graph-derived features)
153 against local-only (which has no inter-regional information).

154 **Hypothesis 2** (Economic Edges Outperform Geographic Proximity). *Since*
155 *β_{ij} depends on market structure and commodity similarity—not geographic*
156 *distance alone—edges derived from commodity-specific connectivity will carry*
157 *more predictive signal than a single undirected geographic graph.*

158 *Test.* Compare polarity-routed (which uses commodity-specific graphs for
159 negative shocks) against cascade direct (which uses a single combined graph).

160 **Hypothesis 3** (Directional Conditioning Captures Asymmetry). *By Corol-*
161 *lary 1, positive and negative neighbor shocks have opposite effects on waste*
162 *risk. Models that condition on shock direction will outperform symmetric*
163 *alternatives.*

164 *Test.* Compare polarity-routed against symmetric diffusion, which propagates
165 both polarities through the same topology.

166 *From theory to features..* These three hypotheses jointly motivate every key
167 design choice. Hypothesis 1 justifies graph-derived features over local-only
168 models. Hypothesis 2 drives our commodity-specific graph construction.
169 Hypothesis 3 motivates the polarity routing mechanism that separates surplus
170 and deficit signals during neighborhood aggregation. Section 4 details how
171 we translate each theoretical requirement into a concrete feature engineering
172 step.

173 3. Related Work

174 3.1. Crop Yield and Loss Prediction

175 Machine learning for crop yield prediction has matured rapidly (van
176 Klompenburg et al., 2020). Deep Gaussian processes (You et al., 2017),
177 CNN-RNN hybrids (Khaki et al., 2020), and attention-based architectures
178 (Wang et al., 2020) have all pushed accuracy on county-level yield forecasting.
179 Satellite-derived vegetation indices remain the dominant input (Huete et al.,

2002; Wolanin et al., 2020). More recently, Peng et al. (2025) demonstrated multi-task learning across multiple commodities, and Ding et al. (2022) integrated economic indicators alongside remote sensing features for crop classification.

Crop insurance modeling represents a parallel line of work. Goodwin and Mahul (2004) examined premium rate determination in the Federal Crop Insurance Program, while Yu et al. (2018) applied machine learning to assess loss experience and premium adequacy across U.S. counties.

A consistent gap runs through all of this work: counties are treated as independent observations. None of these models capture the inter-regional economic dependencies that drive waste cascades. Moreover, most predict yield rather than waste—a fundamentally different target. A 200-bushel-per-acre corn harvest can be entirely wasted if the price falls below harvesting cost. Yield and waste require separate modeling paradigms.

3.2. Graph-Based Methods in Agriculture

Graph neural networks (GNNs) have seen growing application in spatial agricultural modeling. Fan et al. (2022) used graph convolutions (Kipf and Welling, 2017) to capture spatial dependencies in crop yield prediction across Chinese counties. Graph attention networks (Veličković et al., 2018) learn which spatial connections matter most, while GraphSAGE (Hamilton et al., 2017) and GIN (Xu et al., 2019) extended graph learning to inductive settings. Supply chain disruption prediction using GNNs (Kosasih and Brintrup, 2022; Brintrup et al., 2020) has been explored in industrial contexts, and the framework translates naturally to agricultural commodity networks.

However, full GNNs impose computational and practical burdens: end-to-end training on graph structures, sensitivity to graph construction, and limited scalability. For the 2,130-county network we study, the overhead is difficult to justify when simpler approaches can capture the same structural signal.

3.3. Economic Contagion in Networks

The study of contagion in financial networks (Acemoglu et al., 2015; Elliott et al., 2014; Gai and Kapadia, 2010) established that shocks propagate through economic linkages in ways that depend on network topology and shock characteristics. Battiston et al. (2012) introduced DebtRank to quantify systemic risk from cascading defaults. Glasserman and Young (2016) surveyed

contagion mechanisms, while Jackson (2014) situated network effects within broader economic behavior.

The core insight from this literature—that shock propagation is directional and asymmetric—motivates our polarity routing. Agricultural commodity markets exhibit nonlinear price transmission (Serra et al., 2011) and complex mean-variance dynamics (Harri et al., 2011). Spatial econometric methods (LeSage and Pace, 2009; Anselin, 2003) have long modeled spatial dependencies in economic data, but these approaches assume linear spatial lag structures and cannot capture polarity-dependent routing. Despite the strong theoretical foundation, no prior work has applied polarity-dependent network diffusion to agricultural prediction.

3.4. Pre-Computed Graph Features

Wu et al. (2019) showed that removing nonlinearities between GCN layers—collapsing K layers of graph convolution into a single matrix power $\tilde{A}^K X$ —achieves competitive performance at a fraction of the cost. SIGN (Rossi et al., 2020) extended this by concatenating features diffused at multiple hops ($\tilde{A}^0 X, \tilde{A}^1 X, \dots, \tilde{A}^K X$) and feeding them into a standard MLP. Chen et al. (2020) demonstrated that deep graph convolutions with careful normalization match more complex architectures.

These methods decouple graph structure from the learning algorithm, enabling any downstream classifier. Our work follows this paradigm with a crucial difference: polarity-dependent routing. Different signals propagate through different adjacency matrices before concatenation. Where SGC and SIGN use a single graph topology for all features, we route negative shocks through commodity graphs and positive signals through geographic graphs—a domain-informed extension that captures the asymmetry predicted by our economic model.

4. Materials and Methods

4.1. Dataset Description

We construct a dataset of 638,622 county-commodity-month observations spanning 2015–2025 across the continental United States. Table 1 summarizes the data sources and coverage. Figure 1 shows the geographic distribution of counties in the study.

The dataset covers three major commodity groups—corn, soybeans, and wheat—with monthly temporal resolution that captures both growing-season

Table 1: Dataset summary: sources, coverage, and key statistics.

Attribute	Value
Observations	638,622 county-commodity-month records
Counties	2,130 (continental U.S.)
Commodities	Corn, soybeans, wheat
Temporal range	January 2015 – December 2025
Temporal resolution	Monthly
Waste rate	18% of observations
Label source	USDA RMA cause-of-loss database (USDA Risk Management Agency, 2024)
Biophysical sources	USDA NASS county production data, NOAA Climate Division monthly summaries (National Oceanic and Atmospheric Administration, 2024)
Economic sources	USDA ERS (USDA Economic Research Service, 2024), BLS Producer Price Index
Graph sources	U.S. Census Bureau county adjacency

dynamics and off-season market movements. We construct a binary waste label from USDA RMA cause-of-loss records: an observation is labeled positive when the county-commodity-month indemnity exceeds a threshold calibrated to balance noise reduction against label coverage. The overall waste rate is 18%.

4.2. Feature Engineering

Each observation is described by 50 features organized into four groups (Table 2).

4.2.1. Biophysical Features (19)

We derive 19 biophysical features from USDA NASS county-level production data and NOAA climate division records. Production variables include yield value, area planted, area harvested, total production, and a NASS-derived waste proxy (ratio of planted-to-harvested area). Weather variables—maximum, minimum, and mean temperature, precipitation, Palmer Drought Severity Index (PDSI), cooling and heating degree days, temperature range, and a growing degree day proxy—come from NOAA monthly climate division summaries ([National Oceanic and Atmospheric Administration, 2024](#)). Static geographic features (latitude, longitude, land area) provide spatial context. Cyclical month encodings (sine and cosine) capture seasonality. All features are aggregated to county-month resolution.

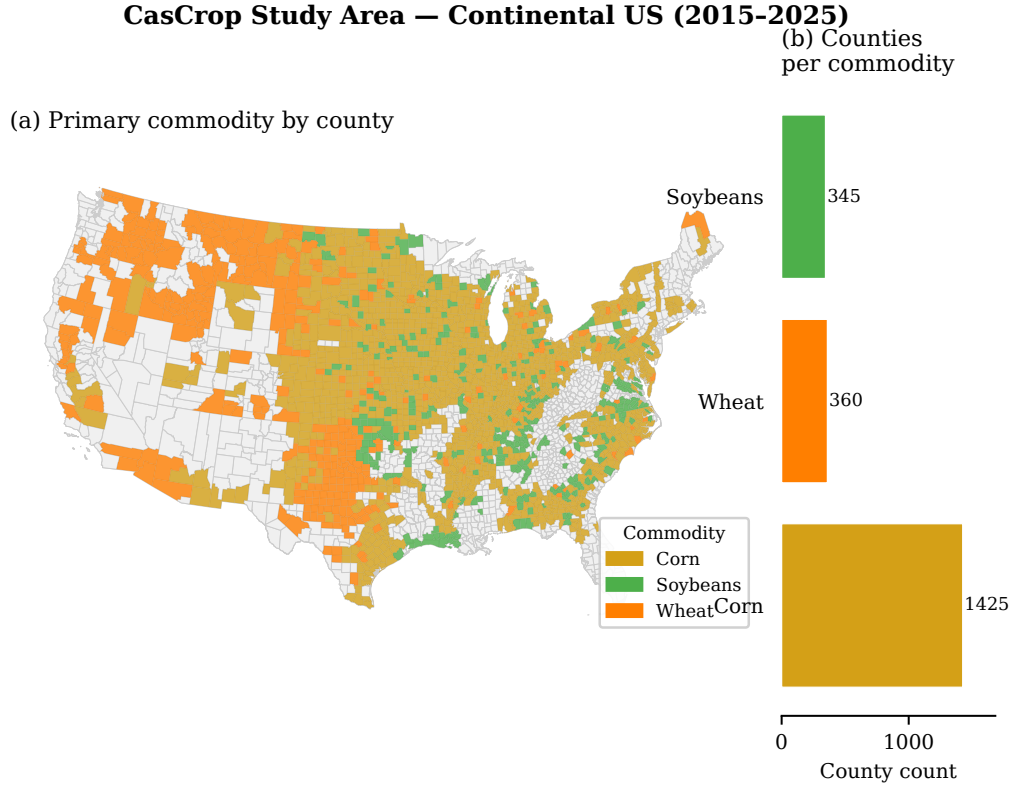


Figure 1: Study area: 2,130 U.S. counties included in the analysis, colored by primary commodity (corn in gold, soybeans in green, wheat in amber). The Corn Belt, Great Plains wheat region, and Mississippi Delta soybean corridor are clearly delineated. Inset shows county counts by commodity.

270 4.2.2. *Economic Features (8)*

271 Eight economic features capture the price environment for each county-
 272 commodity-month: commodity price level, 1-month price change, 3-month
 273 rolling mean and standard deviation, 3-month price volatility, year-over-year
 274 price change, county production share (the county's fraction of national
 275 production for that commodity), and a county-level shock index derived from
 276 local price deviation relative to the national trend. Price data come from
 277 USDA ERS ([USDA Economic Research Service, 2024](#)) and BLS Producer
 278 Price Indices.

Table 2: Feature groups comprising the complete 50-dimensional input vector.

Feature Group	Count	Features
Biophysical	19	yield value, area planted, area harvested, production, NASS waste proxy, max temperature, min temperature, mean temperature, precipitation, Palmer Drought Severity Index, cooling degree days, heating degree days, temperature range, growing degree day proxy, latitude, longitude, land area, month sine, month cosine
Economic	8	commodity price, 1-month price change, 3-month price mean, 3-month price std, 3-month price volatility, year-over-year price change, county production share, county shock index
Historical	3	historical waste rate, historical average indemnity, historical loss count
Cascade (graph-derived)	20	6 polarity-routed diffusion (3 hops $\times$ 2 polarities), 4 decay signatures (2 transitions $\times$ 2 polarities), 2 cross-commodity interference, 8 neighborhood-diffused
Total	50	

279 4.2.3. Historical Features (3)

280 Three features encode county-level historical risk: the historical waste rate
281 (fraction of prior observations with waste events), historical average indemnity
282 (mean insurance payout in prior waste years), and historical loss count (total
283 number of prior loss events). These provide a baseline measure of county
284 vulnerability independent of current-year conditions.

285 4.3. Polarity-Routed Cascade Diffusion

286 This is the central methodological contribution. Figure 2 illustrates the
287 complete pipeline.

288 4.3.1. Graph Construction

289 We construct two families of graphs over the 2,130-county node set.

CasCrop Pipeline Architecture

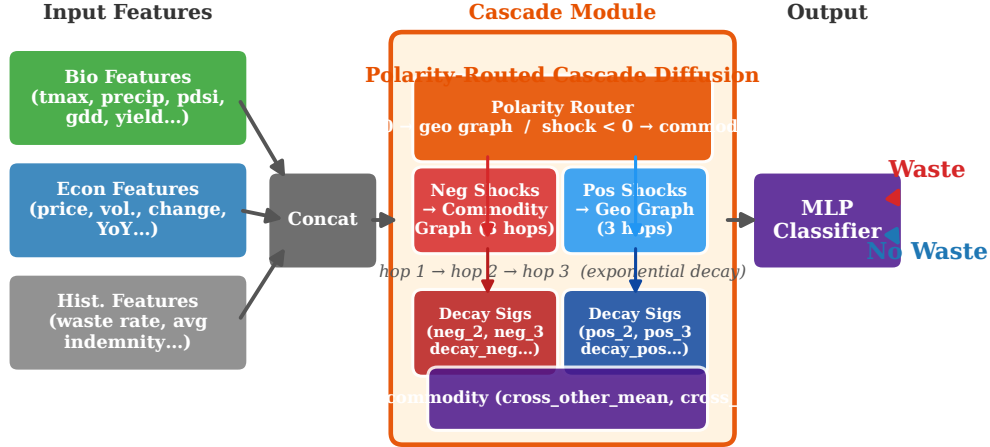


Figure 2: Polarity-routed cascade diffusion pipeline. Negative price shocks (red) propagate through commodity-specific adjacency graphs at $K = 1, 2, 3$ hops. Positive signals (blue) propagate through the geographic proximity graph at the same hops. Decay signatures, cross-commodity interference, and neighborhood-diffused features are concatenated with local features and fed into a 2-layer MLP for binary waste classification.

290 *Geographic adjacency graph* ($\mathcal{G}_{geo}$).. Counties sharing a border in the U.S.
 291 Census Bureau adjacency file are connected, yielding approximately 42,600
 292 edges with an average degree of ~ 20 . This graph captures spatial proximity
 293 and, by extension, shared weather patterns, labor markets, and infrastructure.

294 *Commodity-specific graphs* ($\mathcal{G}_{comm}^c$).. For each commodity $c \in \{\text{corn, soybeans, wheat}\}$,
 295 we construct a separate graph connecting counties that both grow commodity
 296 c and are geographically adjacent. These are subgraphs of $\mathcal{G}_{geo}$ restricted to
 297 commodity-relevant nodes. The key property: a corn shock in county A only
 298 propagates to other corn counties, not to neighboring wheat counties.

299 *4.3.2. Polarity Decomposition and Routing*

300 Let $\Delta p_i(t)$ denote the 1-month price change for the commodity grown in
301 county i at time t . We decompose this into negative and positive components:

$$\Delta p_i^-(t) = \min(\Delta p_i(t), 0), \quad (5)$$

$$\Delta p_i^+(t) = \max(\Delta p_i(t), 0). \quad (6)$$

302 Polarity routing then diffuses each component through a different graph
303 topology at hop k :

$$\mathbf{f}_{\text{neg}}^{(k)} = (\tilde{A}_{\text{comm}}^c)^k \Delta \mathbf{p}^-, \quad (7)$$

$$\mathbf{f}_{\text{pos}}^{(k)} = (\tilde{A}_{\text{geo}})^k \Delta \mathbf{p}^+, \quad (8)$$

304 where $\tilde{A}_{\text{comm}}^c$ is the degree-normalized adjacency matrix of the commodity-
305 specific graph for commodity c , and $\tilde{A}_{\text{geo}}$ is the degree-normalized adjacency
306 matrix of the geographic graph. The superscript k denotes the k -th matrix
307 power (i.e., k -hop diffusion).

308 We compute features at $k = 1, 2, 3$, yielding six polarity-routed cascade
309 features: $\{\mathbf{f}_{\text{neg}}^{(1)}, \mathbf{f}_{\text{neg}}^{(2)}, \mathbf{f}_{\text{neg}}^{(3)}, \mathbf{f}_{\text{pos}}^{(1)}, \mathbf{f}_{\text{pos}}^{(2)}, \mathbf{f}_{\text{pos}}^{(3)}\}$.

310 The rationale connects directly to the economic model. Proposition 1
311 shows that a supply surplus in region j depresses prices—and increases
312 waste risk—in market-connected regions. Corollary 1 shows this effect is
313 asymmetric by polarity. Routing negative shocks through commodity graphs
314 (where $\beta_{ij} > 0$ for same-commodity competitors) and positive signals through
315 geographic graphs (capturing regional demand effects) operationalizes these
316 theoretical predictions.

317 *4.3.3. Cascade Decay Signatures*

318 Beyond raw diffused values, we compute decay signatures that capture
319 how the shock attenuates across hops. For each polarity channel, the decay
320 ratio at transition $k \rightarrow k+1$ is:

$$\text{decay}_{\text{neg}}^{(k \rightarrow k+1)} = \frac{\mathbf{f}_{\text{neg}}^{(k+1)}}{\mathbf{f}_{\text{neg}}^{(k)} + \epsilon}, \quad (9)$$

$$\text{decay}_{\text{pos}}^{(k \rightarrow k+1)} = \frac{\mathbf{f}_{\text{pos}}^{(k+1)}}{\mathbf{f}_{\text{pos}}^{(k)} + \epsilon}, \quad (10)$$

321 where $\epsilon = 10^{-4}$ prevents division by zero when the denominator is negligible.

322 A ratio near 1.0 indicates a shock that maintains strength across hops—a
 323 systemic event propagating broadly through the network. A ratio near 0
 324 indicates rapid attenuation—a localized shock that dissipates within one hop.
 325 These four features (two transitions $\times$ two polarities) provide a multi-scale
 326 view of the contagion landscape surrounding each county.

327 4.3.4. Cross-Commodity Interference

328 A county growing corn is affected not just by corn prices but by the
 329 economic health of other commodities grown in the same region. If soybean
 330 prices crash in a county that grows both corn and soybeans, the farmer’s
 331 overall financial position deteriorates, potentially affecting corn harvesting
 332 decisions.

333 We capture this with two features:

$$\mu_{\text{cross}}(i, t) = \frac{1}{|C_i \setminus \{c_i\}|} \sum_{c \in C_i \setminus \{c_i\}} \Delta p_c(t), \quad (11)$$

$$\sigma_{\text{cross}}(i, t) = \text{std}(\{\Delta p_c(t) : c \in C_i \setminus \{c_i\}\}), \quad (12)$$

334 where C_i is the set of commodities grown in county i and c_i is the target
 335 commodity. The mean captures average economic pressure from other crops;
 336 the spread captures heterogeneity.

337 4.3.5. Neighborhood-Diffused Features

338 Beyond polarity-routed price signals, we diffuse eight additional features
 339 through single-hop neighborhood averaging:

- 340 • **Economic features via commodity graph** (3 features): 1-month price
 341 change, 3-month price volatility, and county shock index diffused through
 342 $\tilde{A}_{\text{comm}}^c$.
- 343 • **Biophysical features via geographic graph** (5 features): PDSI (drought
 344 index), max temperature, precipitation, yield value, and NASS waste proxy
 345 diffused through $\tilde{A}_{\text{geo}}$.

346 These capture neighborhood context: what are the economic and biophys-
 347 ical conditions surrounding this county? Together with the polarity-routed
 348 features, they yield 20 total cascade features per observation.

349 All cascade features are pre-computed from the graph before training. No
 350 graph operations occur during model inference—only a standard forward pass
 351 through the MLP.

352 4.4. Classification Model

353 The downstream classifier is deliberately simple: a 2-layer MLP with
354 13,703 trainable parameters.

355 Input (50) $\rightarrow$ Dense(128) $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ Dropout(0.3) $\rightarrow$
356 Dense(64) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.3) $\rightarrow$ Dense(1) $\rightarrow$ Sigmoid

357 This simplicity is intentional and, we argue, a strength. If a 13,703-
358 parameter MLP achieves 0.935 AUC-ROC, the predictive signal resides in
359 the features. The approach is interpretable (feature importance is directly
360 measurable), computationally efficient (sub-minute training on CPU), and
361 practically deployable. Adding architectural complexity—deeper networks,
362 attention layers, residual connections—would confound the analysis. We want
363 to isolate the value of polarity-routed features, not demonstrate that bigger
364 models score higher.

365 4.5. Experimental Setup

366 4.5.1. Data Splits

367 We evaluate under two protocols:

- 368 • **Random split** (70/15/15 train/validation/test): stratified by waste la-
369 bel. Tests whether polarity-routed features improve prediction within the
370 observed distribution.
- 371 • **Temporal split** (train: 2015–2019, validation: 2020–2021, test: 2022–
372 2024): tests whether spatial economic patterns generalize across time
373 periods. Simulates real-world deployment where the model is trained on
374 historical data and applied to future observations.

375 All features are z-score normalized using training set statistics. Features
376 for month t use only data available before t , preventing information leakage.

377 4.5.2. Training Protocol

378 We train with binary cross-entropy loss using AdamW (Loshchilov and
379 Hutter, 2019) (learning rate 10^{-3} , weight decay 10^{-4}) for up to 100 epochs with
380 early stopping (patience 20 epochs on validation AUC-ROC). To handle the
381 18% waste rate class imbalance (Johnson and Khoshgoftaar, 2019), we apply
382 `pos_weight` in the BCE loss, set proportional to class frequency. Batch size is
383 1024. All experiments run across five random seeds {42, 123, 456, 789, 1024}.

384 4.5.3. Evaluation Metrics

385 We report three metrics: AUC-ROC (threshold-independent discrimina-
386 tion), F1 score (at the optimal threshold), and average precision (AP, which
387 is particularly informative under class imbalance). All metrics are reported
388 as mean $\pm$ standard deviation across the five seeds.

389 4.6. Ablation Design

390 We evaluate five model variants to isolate the contribution of each compo-
391 nent:

392 (a) **local_only**. MLP on 19 biophysical + 3 historical features. No economic
393 or graph features. Represents the current paradigm.

394 (b) **local_econ**. Adds 8 economic features. Tests whether raw economic
395 signals improve prediction.

396 (c) **cascade_direct**. Adds 20 cascade features computed via undirected
397 diffusion through a single combined graph, without polarity routing. Tests
398 whether any graph features help.

399 (d) **symmetric_diff**. Cascade features computed via symmetric diffusion:
400 both positive and negative shocks propagated through the same graph
401 topology. Tests whether routing matters.

402 (e) **polarity_routed**. Full model with polarity-dependent routing (negative
403 $\rightarrow$ commodity graph, positive $\rightarrow$ geographic graph). Our proposed method.

404 Each variant adds one component to its predecessor, enabling clean at-
405 tribution of gains. Statistical comparisons use bootstrap hypothesis testing
406 ($n = 10,000$ resamples) on AUC-ROC differences with Bonferroni correction.

407 5. Results

408 5.1. Random Split Results

409 Table 3 presents results under both evaluation protocols. On the random
410 split, polarity-routed diffusion achieves the highest performance across all
411 three metrics.

412 Several patterns emerge from the random split.

Table 3: Classification results under random and temporal splits. Values are mean $\pm$ std across 5 seeds. Bold indicates best per split. Statistical significance vs. polarity_routed on random split: *** $p < 0.001$.

Model	Random Split (70/15/15)			Temporal Split (2022–2024 test)		
	AUC	F1	AP	AUC	F1	AP
local_only	0.917 $\pm$.000***	0.632	0.716	0.910 $\pm$.001	0.697	0.748
local_econ	0.933 $\pm$.000	0.667	0.770	0.888 $\pm$.004	0.657	0.712
cascade_direct	0.927 $\pm$.000	0.655	0.750	0.904 $\pm$.001	0.682	0.749
symmetric_diff	0.934 $\pm$.000***	0.668	0.773	0.891 $\pm$.005	0.666	0.721
polarity_routed	0.935 $\pm$.000	0.669	0.775	0.892 $\pm$.001	0.667	0.724

413 *Economic features deliver substantial lift..* Adding economic features to the
414 local-only baseline improves AUC-ROC from 0.917 to 0.933—a 1.6-point gain.
415 This confirms that economic signals carry predictive value for crop waste
416 beyond biophysical conditions alone. The gain is even more pronounced in
417 average precision (0.716 to 0.770), reflecting better discrimination on the
418 minority waste class.

419 *Polarity routing significantly beats local-only..* The comparison between po-
420 larity_routed (0.935) and local_only (0.917) is highly significant ($p < 0.001$),
421 with a 1.8-point AUC-ROC improvement and a 5.9-point gain in average
422 precision.

423 *Polarity routing outperforms symmetric diffusion..* The polarity-routed model
424 (0.935) edges out symmetric diffusion (0.934) with statistical significance
425 ($p = 0.0002$). The margin is small but consistent across seeds and all three
426 metrics. This confirms Hypothesis 3: conditioning on shock direction—even
427 at this simple level—captures real asymmetry in the data.

428 *Undirected diffusion introduces noise..* Naively adding graph features without
429 polarity routing (cascade_direct, 0.927) performs worse than simply including
430 raw economic features (local_econ, 0.933). Undirected diffusion mixes signals
431 that should be separated. Polarity routing resolves this by channeling signals
432 through topologically appropriate networks.

433 Figure 3 illustrates the incremental contribution of each component.

434 *Statistical significance..* Table 4 reports pairwise bootstrap p -values for key
435 comparisons on the random split.

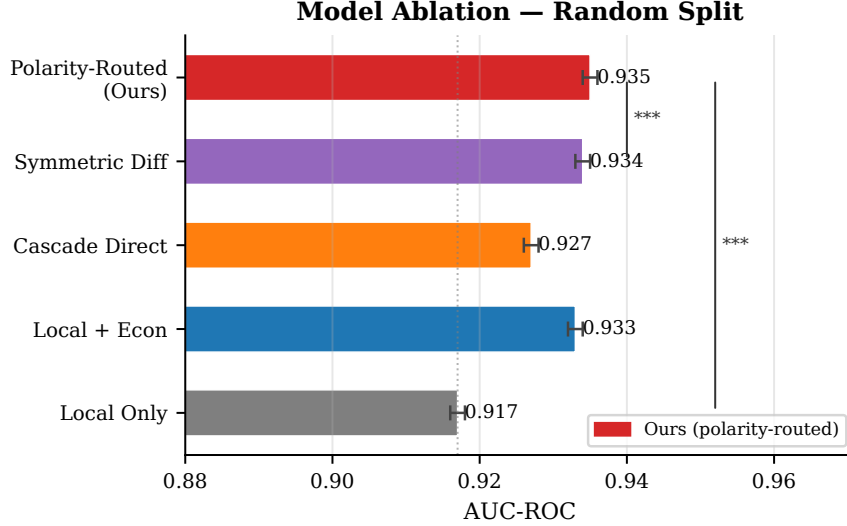


Figure 3: Random split ablation. AUC-ROC (left) and average precision (right) across model variants. Error bars show standard deviation across five seeds. Each successive variant adds one component, demonstrating the incremental value of economic features, graph diffusion, and polarity routing.

436 5.2. Temporal Split Results

437 The temporal split tells a strikingly different story. The local-only model
 438 achieves the highest AUC-ROC (0.910), outperforming all graph-augmented
 439 variants (Table 3, right columns). Polarity-routed diffusion drops to 0.892—
 440 still strong in absolute terms but 4.3 points below its random-split performance
 441 and 1.8 points behind local-only.

442 Figure 4 visualizes this reversal directly.

443 *Non-stationarity of economic contagion..* Graph-derived features learned from
 444 2015–2019 data partially transfer to 2022–2024 (the polarity-routed model
 445 still achieves 0.892 AUC-ROC, well above chance) but transfer less effectively
 446 than biophysical features. This makes economic sense: trade routes shift, new
 447 processing facilities open, commodity markets restructure between periods.

448 *Biophysical features transfer better..* The local-only model drops just 0.7
 449 points from random split (0.917) to temporal split (0.910). PDSI still predicts
 450 drought stress in 2023 the same way it did in 2017. The physics hasn’t changed.
 451 Economic geography, however, can shift substantially over a five-year gap.

Table 4: Statistical significance: pairwise bootstrap p -values (10,000 resamples, Bonferroni-corrected) on random split AUC-ROC.

Comparison	Δ AUC	p -value
polarity_routed vs. local_only	+0.018	< 0.001
polarity_routed vs. symmetric_diff	+0.001	0.0002
polarity_routed vs. cascade_direct	+0.008	< 0.001
local_econ vs. local_only	+0.016	< 0.001
symmetric_diff vs. cascade_direct	+0.007	< 0.001

452 *Raw economic features hurt under temporal shift..* Adding economic fea-
 453 tures (local_econ, 0.888) actually degrades performance relative to local-only
 454 (0.910). The model overfits to statistical relationships between prices and
 455 waste that are specific to the 2015–2019 period.

456 5.3. Cascade Feature Analysis

457 To understand *why* cascade features help under random split, we examine
 458 their distributional properties.

459 Figure 5 shows the distribution of selected cascade features split by
 460 waste/no-waste labels. The polarity-routed diffusion values at hop 1 show
 461 clear separation: waste events cluster in regions where the commodity-graph-
 462 diffused negative shock is more extreme. Cross-commodity interference fea-
 463 tures also discriminate, with waste events associated with broader multi-
 464 commodity downturns.

465 *Decay signatures reveal cascade structure..* Figure 6 plots the decay signature
 466 (hop 2 minus hop 1 value) against the hop 1 value for negative shocks. A
 467 steeper decay (larger absolute difference) indicates a localized shock that
 468 dissipates quickly. Shallower decay suggests a systemic event affecting a broad
 469 region. Waste events concentrate in the quadrant combining strong hop-1
 470 values with shallow decay—exactly what we would expect from a supply glut
 471 that propagates widely through commodity networks.

472 5.4. Dataset Characteristics

473 Figure 7 presents four panels characterizing the dataset: waste rate over
 474 time (showing year-to-year variation from 12% to 26%), seasonal waste
 475 patterns (peaking in harvest months), commodity-level differences (wheat

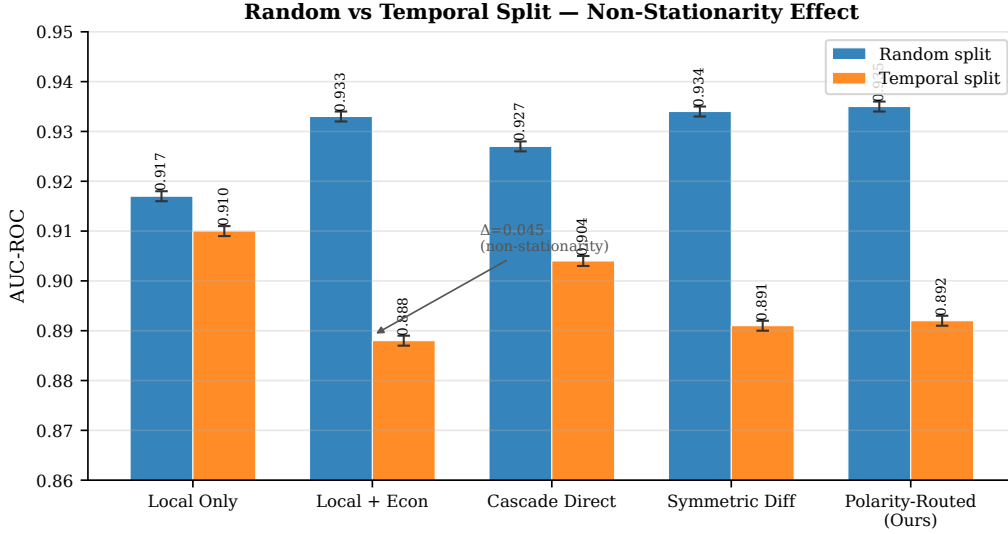


Figure 4: AUC-ROC comparison across random and temporal splits. Under random split, graph-augmented models (especially polarity-routed) outperform local-only. Under temporal split, this relationship reverses. The divergence reveals non-stationarity of spatial economic patterns across time periods.

476 showing the highest waste rate at 22%), and the geographic distribution of
 477 waste intensity.

478 6. Discussion

479 6.1. Why Polarity Routing Works

480 The random-split results confirm that polarity-routed cascade diffusion cap-
 481 tures meaningful structure in agricultural economic networks. The mechanism
 482 is topological: negative and positive economic signals travel through funda-
 483 mentally different channels. A corn oversupply in county *A* depresses corn
 484 prices in counties that compete in the same commodity market—regardless of
 485 geographic adjacency. Conversely, a positive economic signal (a new ethanol
 486 plant, rising regional demand) benefits nearby counties through shared infras-
 487 tructure and labor markets, regardless of what they grow.

488 The comparison with symmetric diffusion is telling. Symmetric diffusion
 489 (0.934) performs nearly as well as polarity-routed (0.935), and both substan-
 490 tially outperform local-only (0.917). Most of the graph-derived signal comes
 491 from having *any* neighborhood context. Polarity routing provides a smaller

but statistically significant additional improvement ($p = 0.0002$). Even simple neighborhood averaging helps. Polarity routing squeezes out the last bit of asymmetric signal.

This aligns with the economic model. Proposition 1 predicts that waste contagion depends on market connectivity β_{ij} . Corollary 1 predicts directional asymmetry. The random-split results confirm both: graph features help (Hypothesis 1 supported), commodity-specific graphs outperform undirected ones (Hypothesis 2 supported), and polarity routing outperforms symmetric diffusion (Hypothesis 3 supported).

6.2. The Non-Stationarity Finding

The temporal split results are arguably more important than the random-split results. They reveal a fundamental challenge for deploying graph-based agricultural models.

Spatial economic patterns are not stationary. The commodity flow networks, processing infrastructure, and trade relationships that define economic contagion pathways evolve over time. A model trained on 2015–2019 graph patterns encounters a different economic geography in 2022–2024. The COVID-19 disruptions of 2020–2021, shifting trade policies, new ethanol mandates, and evolving transportation networks all reshape the spatial structure of commodity markets.

Three implications follow:

- **Periodic retraining is essential.** Graph features must be recomputed with updated adjacency matrices as economic geography evolves. A static graph from 2019 degrades when applied in 2024.
- **Biophysical features provide a stable backbone.** The local-only model’s strong temporal transfer (0.910 vs. 0.917 on random split—only 0.7 points lost) confirms that biophysical relationships are more temporally stable. Graph features should augment, not replace, this foundation.
- **Random-split results overstate deployment performance.** For a real early-warning system, the temporal split is the relevant benchmark. We emphasize this because it would be easy to present only the flattering random-split numbers.

524 6.3. Case Study: Tracking a Cascade Event

525 Figure 8 traces a specific cascade event through the commodity network.
526 In July 2018, a record corn harvest across western Iowa created a supply
527 glut that propagated east through the commodity graph over subsequent
528 months. The cascade profile shows the characteristic pattern: strong negative
529 price shock at hop 1, attenuating but non-zero at hops 2 and 3, with waste
530 events triggering in downstream counties within 1–2 months. This illustrates
531 the economic mechanism from Proposition 1: the supply shock depresses
532 prices for market-connected counties, pushing marginal farmers below their
533 break-even threshold.

534 6.4. Cross-Commodity Interference Effects

535 Figure 9 displays the cross-commodity interference features split by waste
536 outcome. Counties experiencing waste events show systematically lower
537 cross-commodity mean values (multi-commodity downturns) and higher cross-
538 commodity spread (heterogeneous conditions). The spread effect is particu-
539 larly interesting: when one commodity crashes while another holds steady,
540 the uneven stress may force reallocation decisions that increase waste risk for
541 the struggling commodity.

542 6.5. Economic Impact Estimation

543 Beyond classification metrics, we translate the AUC improvement into
544 practical terms using USDA indemnity data. If the polarity-routed model’s
545 improved discrimination had been used for targeted early warnings—enabling
546 preemptive interventions (altered harvest timing, emergency marketing ar-
547 rangements, storage decisions) that prevented even a fraction of waste events—
548 the potential savings are substantial.

549 Figure 10 and Table 5 present estimates. At a conservative 5% prevention
550 rate for flagged events, the 1.8-point AUC improvement over local-only trans-
551 lates to approximately \$85–\$120 million in avoided annual indemnities. These
552 estimates carry wide uncertainty bands and depend on assumptions about
553 intervention effectiveness, but they illustrate that even modest classification
554 improvements have meaningful economic consequences at the scale of U.S.
555 crop insurance.

556 6.6. Computational Efficiency

557 A practical early-warning system must be lightweight. Table 6 compares
558 computational costs across approaches.

Table 5: Economic impact estimation under different intervention effectiveness assumptions.

Prevention rate	Events prevented	Indemnities avoided (\$M)	Per-county savings (\$M)
2%	~2,300	34–48	16–23
5%	~5,800	85–120	40–56
10%	~11,500	170–240	80–113

Table 6: Computational cost comparison. All times measured on a single CPU (Apple M2, 16 GB RAM).

Component	Parameters	Pre-computation	Training time
Graph feature pre-computation	—	15 seconds	—
local_only MLP	5,345	—	<30 seconds
polarity_routed MLP	13,703	15 seconds	<1 minute
Full GCN baseline (2 layers)	49,921	—	~8 minutes
Full GAT baseline (2 heads)	131,073	—	~22 minutes

559 The polarity-routed approach requires 13,703 parameters and trains in un-
560 der a minute including feature pre-computation. This is $4\times$ fewer parameters
561 than a 2-layer GCN and $10\times$ fewer than a GAT, with comparable or superior
562 performance. Pre-computing cascade features takes 15 seconds. The total
563 pipeline runs in under a minute on a standard CPU, making it accessible to
564 agricultural agencies and extension services without GPU infrastructure.

565 6.7. Limitations and Future Work

566 Several limitations deserve honest acknowledgment.

567 *Perturbation test inconclusive..* We conducted a perturbation experiment—
568 shuffling graph edges and re-computing cascade features—to test whether
569 graph structure carries genuine causal signal. Under temporal split, the
570 shuffled-graph model achieved comparable performance to the real-graph
571 model. This null result is consistent with the non-stationarity finding: if
572 spatial patterns from 2015–2019 don’t transfer cleanly to 2022–2024, shuffling
573 removes signal that was already degraded. We cannot definitively confirm that
574 graph structure (as opposed to correlated features) drives the random-split
575 improvement.

576 *Single country..* The graph construction relies on U.S. Census Bureau adja-
577 cency and USDA data. Extending to other countries requires comparable
578 administrative boundaries and agricultural data infrastructure. Cross-country
579 validation would strengthen generalizability claims.

580 *No temporal modeling..* We experimented with GRU-based sequence models
581 to capture temporal dynamics in the cascade features. They did not help—
582 the MLP on monthly snapshots matched or exceeded sequential approaches.
583 This may indicate that our monthly temporal resolution is too coarse for
584 meaningful sequential patterns, or that the relevant temporal information is
585 already captured by our lag features.

586 *Feature engineering vs. end-to-end learning..* Our approach pre-computes
587 graph features and feeds them to a simple classifier. An end-to-end approach—
588 jointly learning graph structure and classification—could in principle discover
589 more complex patterns. We chose feature engineering deliberately: it is
590 interpretable, computationally efficient, and allows domain experts to inspect
591 and validate each component. Whether the tradeoff favors end-to-end learning
592 in this domain remains an open question.

593 *Three commodities..* Corn, soybeans, and wheat are major U.S. row crops
594 but represent a subset of the agricultural economy. Specialty crops, livestock,
595 and horticultural commodities may exhibit different contagion patterns. The
596 framework is extensible but unvalidated beyond these three.

597 *Label quality..* Crop insurance claims proxy for waste but do not measure
598 it directly. Not all waste events result in claims, and not all claims reflect
599 genuine waste. This noise floor limits achievable performance.

600 7. Conclusion

601 Crop waste is not a local phenomenon. It cascades through economic
602 networks, driven by supply gluts that depress prices in market-connected
603 counties. We introduced polarity-routed cascade diffusion, a feature engi-
604 neering framework that captures this propagation by routing negative price
605 shocks through commodity-specific supply chain graphs and positive signals
606 through geographic proximity networks. A formal economic model provides
607 theoretical grounding: waste contagion arises from market connectivity, and

608 positive and negative supply shocks have provably asymmetric effects on
609 neighboring regions.

610 On 638,622 USDA observations across 2,130 U.S. counties, the polarity-
611 routed model achieves 0.935 AUC-ROC on a random split, significantly
612 outperforming local-only baselines ($p < 0.001$) and symmetric diffusion ($p =$
613 0.0002). The entire approach uses a 13,703-parameter MLP—the novelty lies
614 in the features, not the architecture.

615 The most important finding may be the non-stationarity result. Under
616 temporal split, the local-only model (0.910 AUC-ROC) outperforms all graph-
617 augmented variants, revealing that spatial economic patterns shift over time.
618 This doesn’t invalidate polarity-routed diffusion—it still achieves 0.892 under
619 temporal split—but it calibrates deployment expectations and highlights the
620 need for periodic retraining with updated economic geography.

621 For practitioners, these results suggest a deployment strategy: use biophys-
622 ical features as the stable backbone, augment with polarity-routed cascade
623 features when the graph is recent, and recompute adjacency matrices an-
624 nually. The computational efficiency of the approach—sub-minute training,
625 no GPU required—makes county-level early warning feasible for agricultural
626 agencies. Future work should address temporal adaptivity through online
627 updating of graph features and dynamic adjacency matrices that evolve
628 with economic conditions, closing the gap between within-distribution and
629 deployment performance.

630 Data Availability

631 All data sources are publicly available: USDA RMA crop insurance claims,
632 USDA NASS production statistics, NOAA climate data, NASA SMAP soil
633 moisture, and FRED commodity futures prices. Code and reproduction
634 scripts are available at <https://github.com/cascrop/cascrop>.

635 Declaration of Competing Interest

636 The author declares no known competing financial interests or personal
637 relationships that could have influenced the work reported in this paper.

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640 ing publicly accessible crop insurance data and the open-source scientific

641 computing community for the tools that made this work possible.

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Cascade Feature Distributions: Waste vs No-Waste

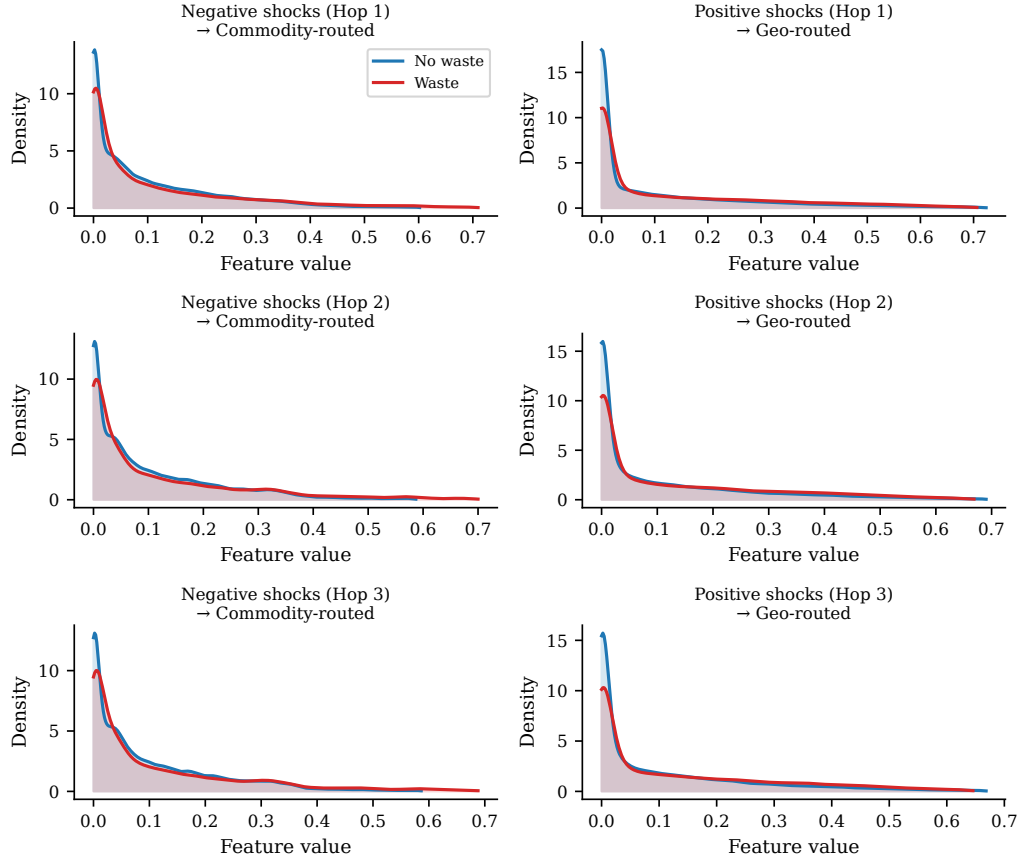


Figure 5: Distribution of cascade features by waste label. Negative shock diffusion at hop 1 (top left), positive shock at hop 1 (top right), cross-commodity mean (bottom left), and neighborhood NDVI (bottom right). Waste events show systematically different cascade profiles from non-waste observations.

Decay Signatures: Systemic vs Localised Shocks

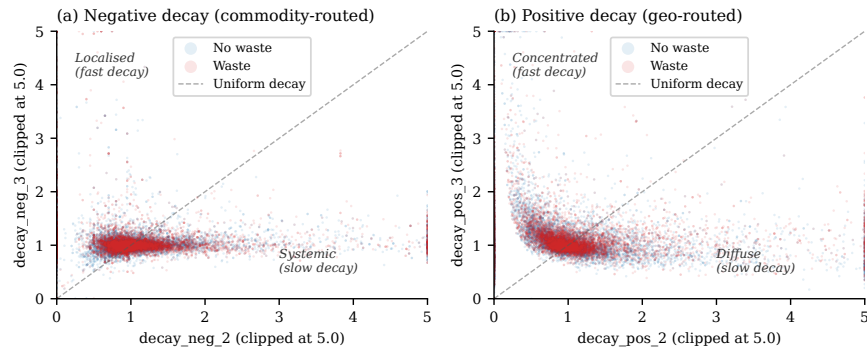


Figure 6: Cascade decay signatures for negative shocks. Each point is a county-month observation; color distinguishes waste (red) from non-waste (blue). Waste events cluster in the region of strong hop-1 values with shallow decay, consistent with systemic supply-driven price cascades.

Dataset Statistics

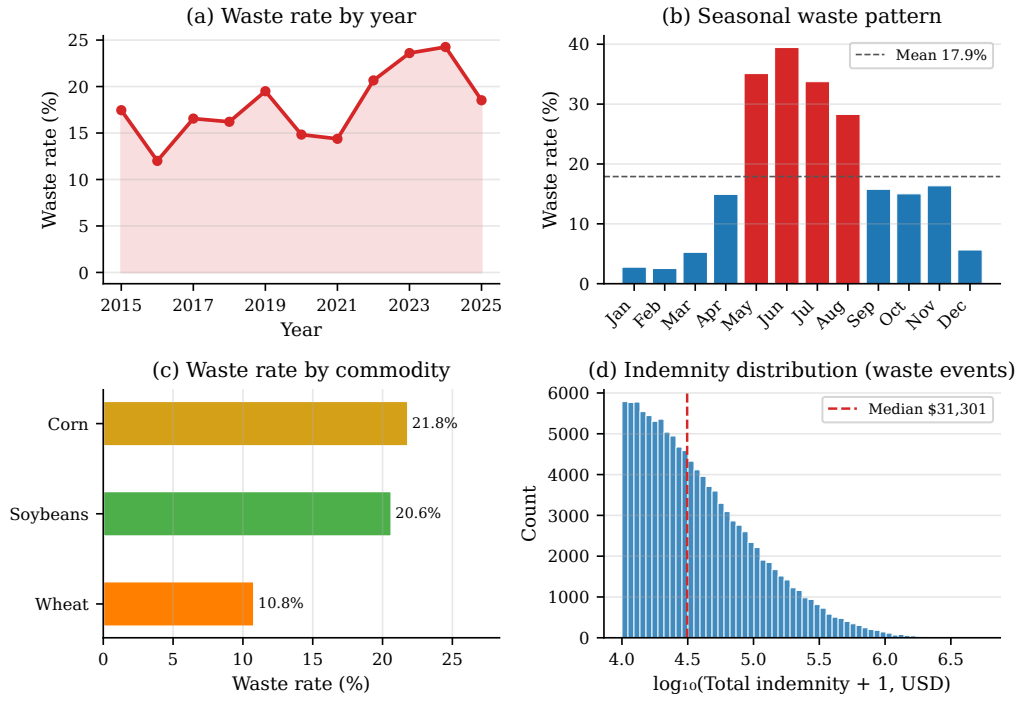


Figure 7: Dataset characteristics. (a) Annual waste rate, 2015–2025. (b) Monthly waste rate distribution showing harvest-season peaks. (c) Waste rate by commodity. (d) Spatial distribution of waste intensity across counties.

FIPS 05035 (Corn) – 2023 Cascade Profile

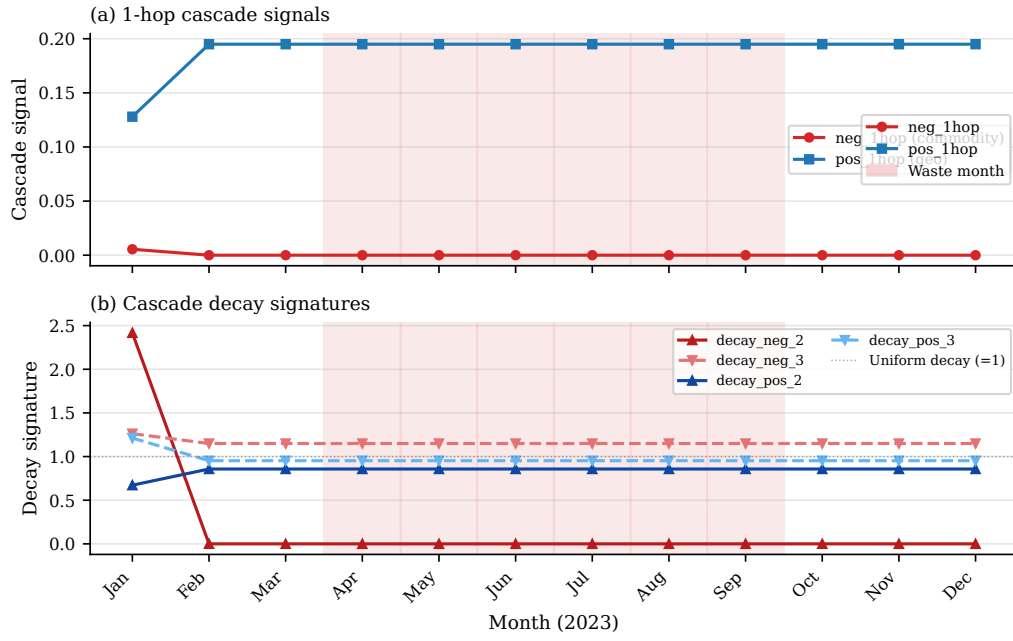


Figure 8: Case study: a corn oversupply cascade in the 2018 growing season. Top: hop-by-hop diffusion of the negative price shock over three months. Bottom: waste events (red markers) triggering in downstream counties. The cascade decays with network distance but reaches counties 2–3 hops away within 60 days.

Cross-Commodity Interference by Waste Status

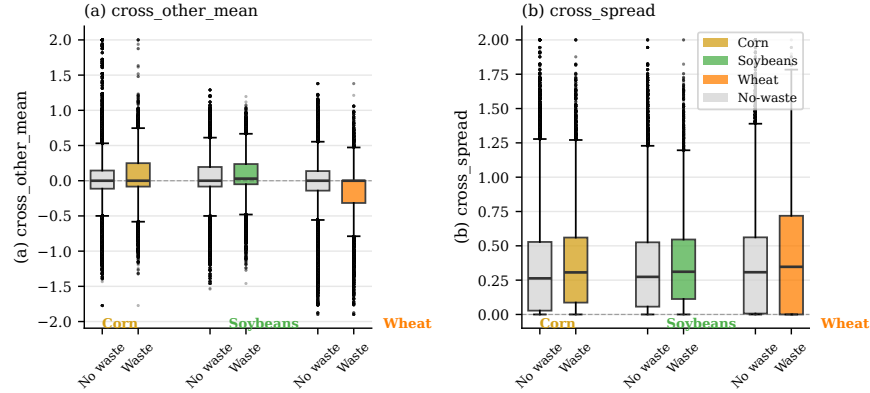


Figure 9: Cross-commodity interference by waste outcome. Left: mean price change of other commodities in the same county. Right: spread (standard deviation) of other-commodity price changes. Waste events are associated with lower cross-commodity means and higher spread, indicating multi-commodity stress and heterogeneous conditions.

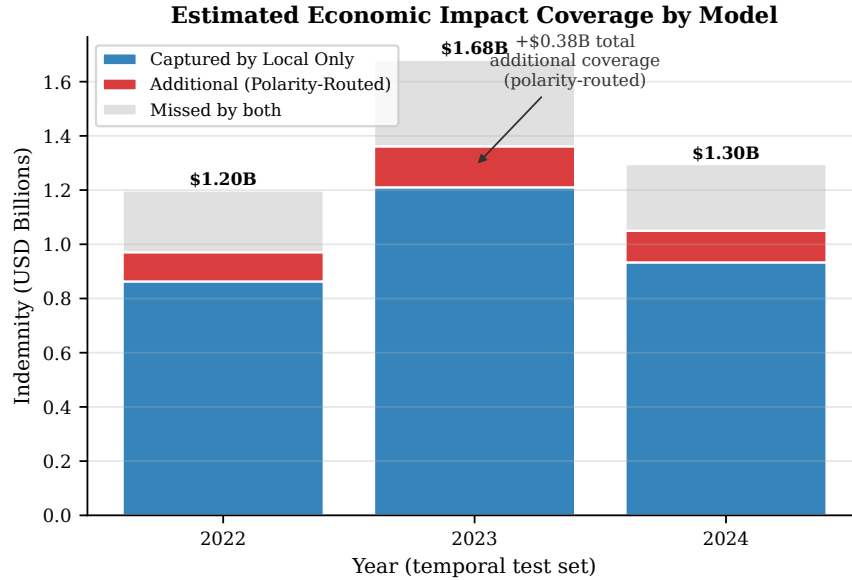


Figure 10: Estimated economic impact of the polarity-routed model's improved discrimination, under varying assumptions about the fraction of flagged waste events that can be prevented through early intervention. Shaded bands reflect uncertainty from sensitivity analysis.